# INTRODUCTION TO INTERNET OF THINGS

MOOC Course Report | NPTEL

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CSE

July - October 2025

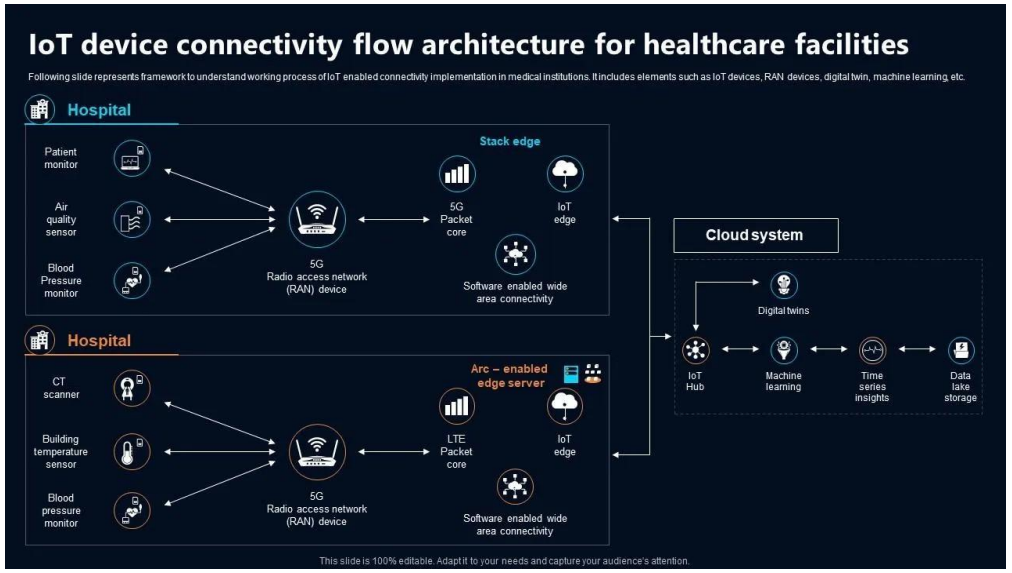
# WHAT IS IOT?

**The Internet of Things (IoT) is a network of physical objects embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.**

### Sensing
Gathering data from the physical environment.

### Connectivity
Transmitting data to the cloud or other devices.

### Processing
Analyzing data to extract meaningful insights.




IOT SYSTEM ARCHITECTURE

# SENSING & ACTUATION

## Sensors

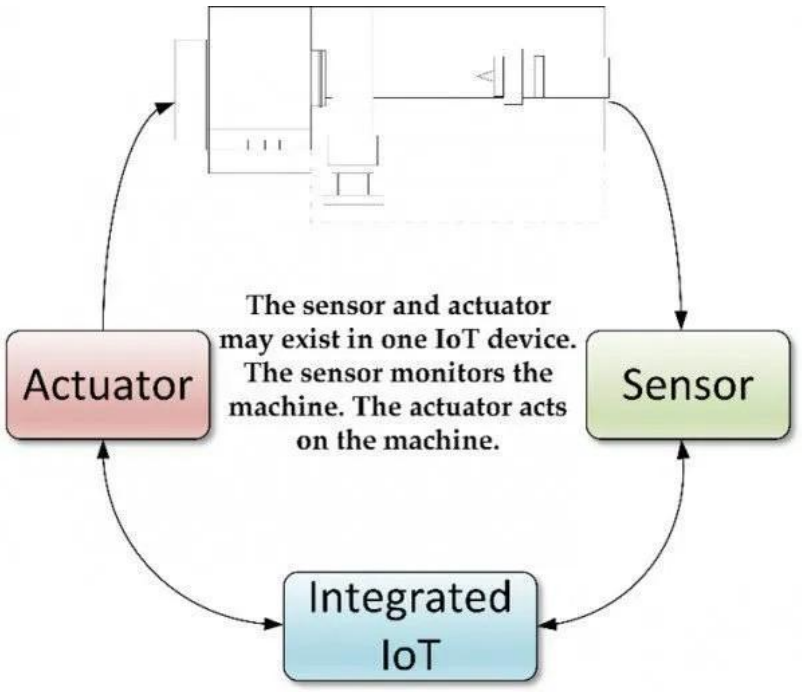
Devices that detect and respond to physical inputs from the environment.

Temperature | Humidity | Motion | Light

## Actuators

Components responsible for moving or controlling a mechanism or system.

Motors | Relays | Valves | Solenoids




SENSOR-ACTUATOR CONTROL LOOP

# NETWORKING & PROTOCOLS

## COMMUNICATION MODELS

### Device-to-Device (D2D)

Direct communication between devices without intermediate cloud services (e.g., Bluetooth, Zigbee).

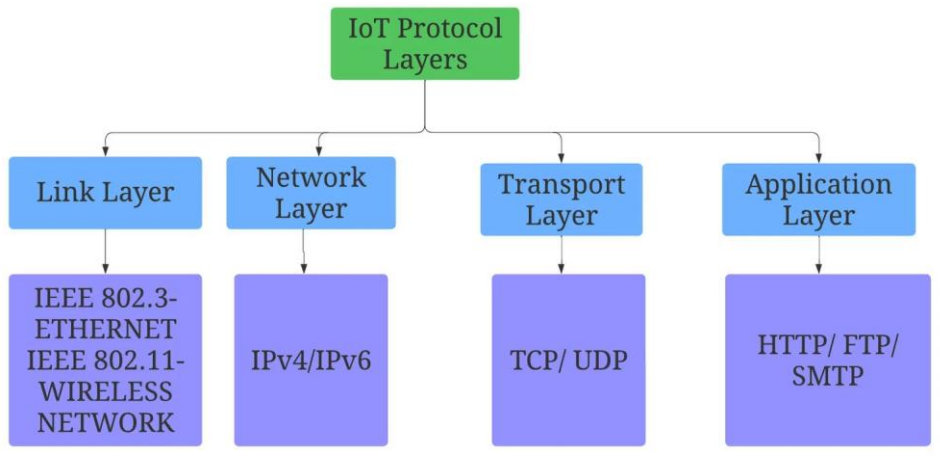
### Device-to-Cloud (D2C)

Devices connect directly to application service providers via IP networks (e.g., Wi-Fi, Ethernet).

### Device-to-Gateway

Devices connect to a local gateway that performs protocol translation and data forwarding.

## IOT PROTOCOL STACK




| LAYER | KEY PROTOCOLS |
|---|---|
| **Application** | MQTT, CoAP, HTTP, AMQP |
| **Transport** | TCP, UDP |
| **Network** | IPv6, 6LoWPAN, RPL |
| **Physical/Link** | IEEE 802.15.4, LoRa, NB-IoT |

# SENSOR NETWORKS & M2M

## SENSOR NETWORKING

Distributed nodes collaborating to monitor environmental conditions. Optimized for large-scale data collection under resource constraints.

- WSN (Wireless)
- BSN (Body Sensor)
- Underwater Nodes
- Mobile Networks

## M2M COMMUNICATION

Automated data exchange between devices without human intervention. M2M provides the autonomous connectivity layer for IoT.

- Point-to-Point
- Remote Monitoring
- Automated Control
- Scalable Architecture

# HARDWARE & PROGRAMMING

## ARDUINO

Microcontroller platform for rapid prototyping and simple sensing/actuation tasks.

- Low Power Consumption
- Real-time Processing
- C/C++ Environment

## PYTHON

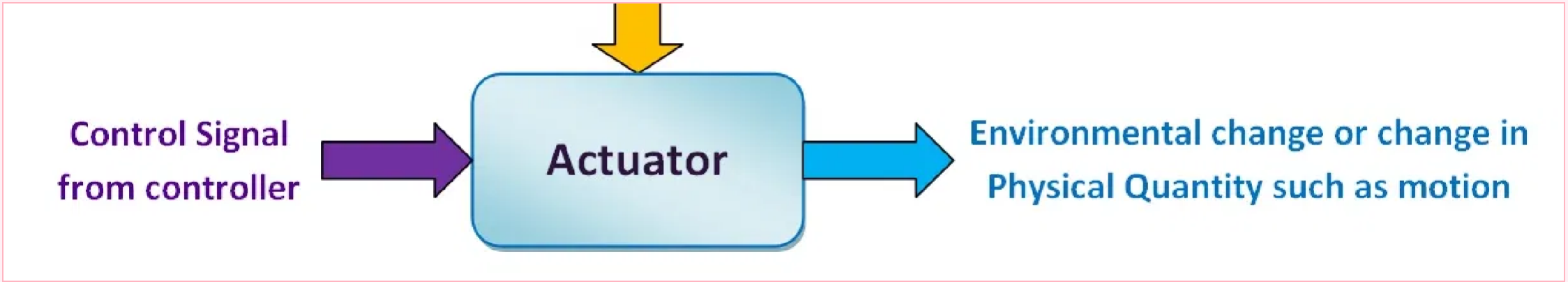
The primary language for IoT due to its simple syntax and extensive library support.

- GPIO Control Libraries
- Data Analytics Ready
- Cloud Integration

## RASPBERRY PI

Single-board computer (SBC) running Linux; acts as a powerful IoT gateway.

- High Processing Power
- Multi-tasking OS
- Built-in Networking



# SDN, ANALYTICS & CLOUD

## SOFTWARE DEFINED NETWORKING

Decouples the control plane from the data plane, enabling programmable and centralized management of complex IoT network traffic.

## CLOUD INFRASTRUCTURE

Provides the elastic storage and computational power required for IoT data handling through IaaS, PaaS, and SaaS models.

## DATA HANDLING & ANALYTICS

Extracts actionable insights from raw sensor streams using advanced analytical frameworks:

DESCRIPTIVE | PREDICTIVE | PRESCRIPTIVE

# SENSOR CLOUD & FOG

## Sensor Cloud

Integration of WSN and Cloud computing to manage distributed sensor data as a scalable service.

## Fog Computing

Extends cloud capabilities to the network edge for localized processing and low-latency response.

### Reduced Latency

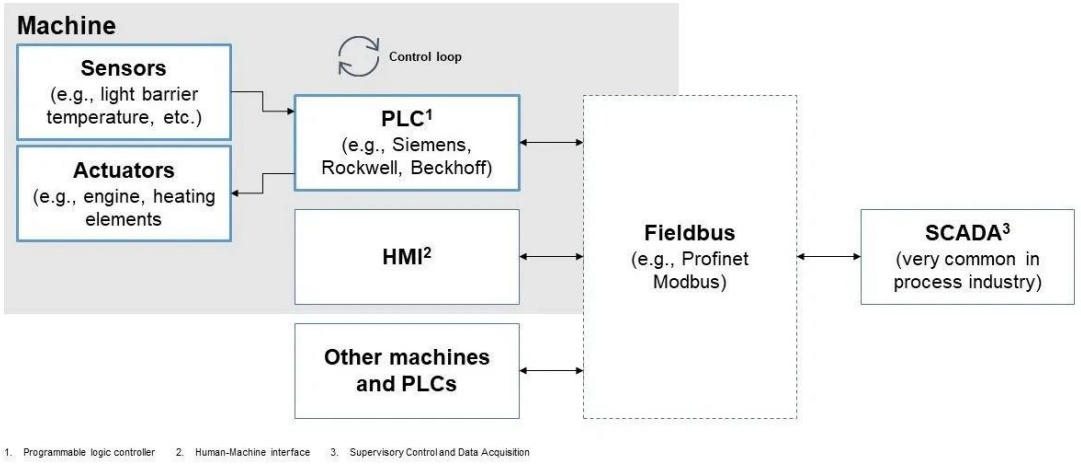
Real-time response for time-critical IoT applications.

### Bandwidth Efficiency

Local data filtering reduces traffic to the central cloud.

### Improved Reliability

Maintains operations during intermittent cloud connectivity.




INDUSTRIAL EDGE PROCESSING ARCHITECTURE

# IOT APPLICATIONS

## SMART CITIES

Intelligent traffic management, waste monitoring, and public safety systems.

## SMART HOMES

Automation of lighting, security, climate control, and smart appliances.

## INDUSTRIAL IOT (IIOT)

Predictive maintenance, smart manufacturing, and supply chain optimization.

## CONNECTED VEHICLES

V2V and V2I communication for enhanced road safety and traffic efficiency.

## SMART GRID

Two-way energy monitoring, demand response, and renewable integration.

# CONCLUSION & FUTURE

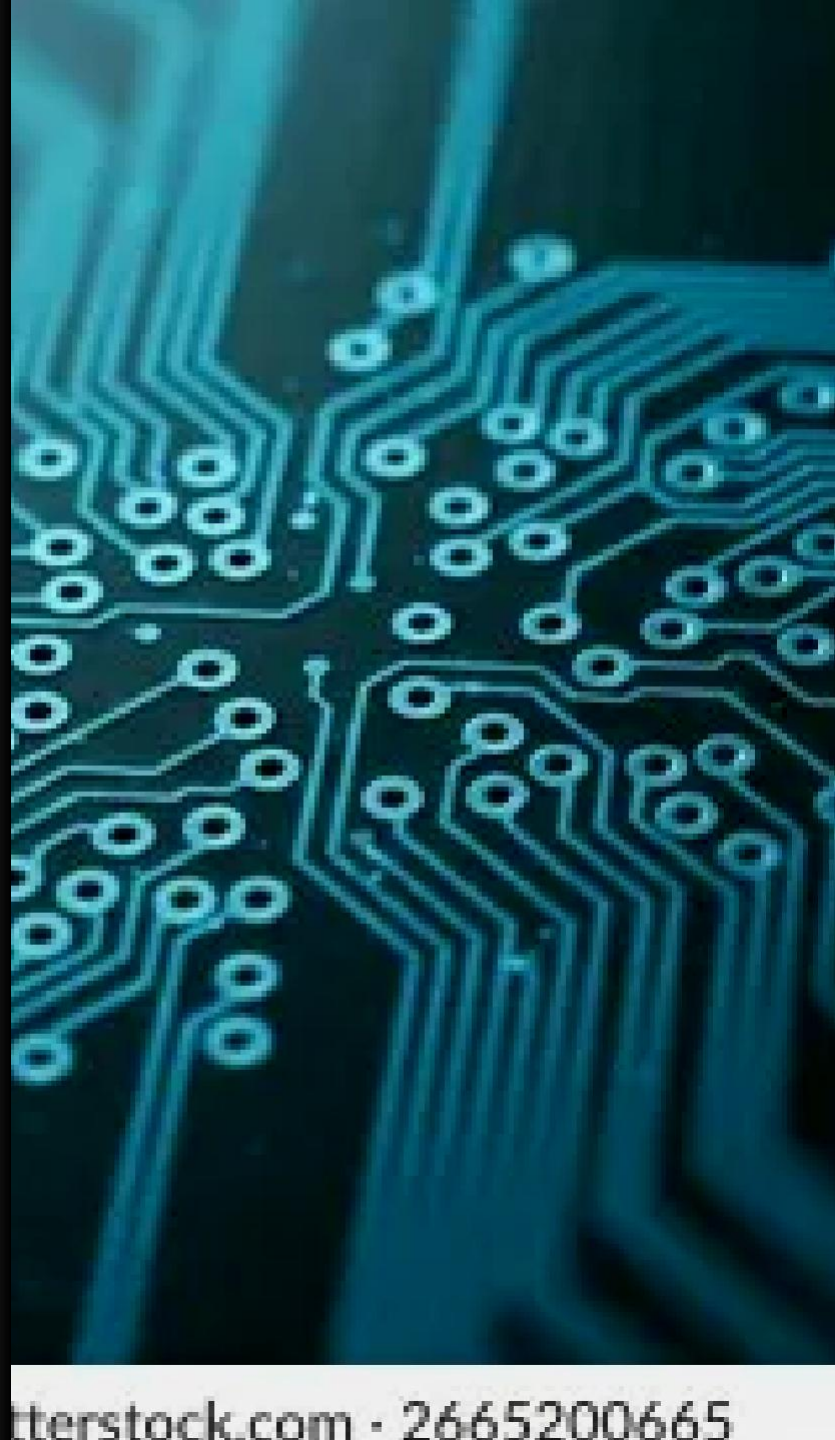
## KEY TAKEAWAYS

- IoT is the foundational technology for digital transformation and smart environments.
- Coordinated sensing, connectivity, and analytics enable real-time intelligent systems.

## CRITICAL CHALLENGES

- Security & Privacy: Protecting massive data streams from evolving cyber threats.
- Interoperability: Achieving seamless communication across heterogeneous devices.
- Scalability: Managing billions of connected nodes with energy efficiency.

## FUTURE SCOPE

- AIoT: Integration of AI for autonomous, self-healing industrial systems.
- Digital Twins: Real-time virtual replicas for predictive simulation.
- 6G Networks: Enabling ultra-low latency and massive machine-type communication.

# THANK YOU